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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Write out the first few terms of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{3^n}$. What is the series' sum?

Evaluate the first term, a_0 , by substituting $n = 0$ in $\frac{(-1)^n}{3^n}$ and using $v^0 = 1$ (any v except $v = 0$).

$$\begin{aligned} a_0 &= \frac{(-1)^n}{3^n} \\ &= \frac{(-1)^0}{3^0} \\ &= 1 \end{aligned}$$

Evaluate the second term, a_1 , by substituting $n = 1$ in $\frac{(-1)^n}{3^n}$ and using $v^1 = v$, (any v).

$$\begin{aligned} a_1 &= \frac{(-1)^n}{3^n} \\ &= \frac{(-1)^1}{3^1} \\ &= -\frac{1}{3} \end{aligned}$$

Evaluate the third term, a_2 , by substituting $n = 2$ in $\frac{(-1)^n}{3^n}$ and using $(-1)^n = 1$, if n is even.

$$\begin{aligned} a_2 &= \frac{(-1)^n}{3^n} \\ &= \frac{(-1)^2}{3^2} \\ &= \frac{1}{9} \end{aligned}$$

Evaluate the fourth term, a_3 , by substituting $n = 3$ in $\frac{(-1)^n}{3^n}$ and using $(-1)^n = -1$, if n is odd.

$$\begin{aligned} a_3 &= \frac{(-1)^n}{3^n} \\ &= \frac{(-1)^3}{3^3} \\ &= -\frac{1}{27} \end{aligned}$$

Compare the given series, $\sum_{n=0}^{\infty} \frac{(-1)^n}{3^n}$, with the general series $a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$. The first term, a , of the series is a_0 from above and is 1.

The constant ratio, r , of consecutive terms is $-\frac{1}{3}$, which may be obtained by dividing the second term, a_1 , by the first term, a_0 , that is, divide $-\frac{1}{3}$ by 1.

The series converges if $|r| < 1$ and the series' sum, s , is given by $\frac{a}{1-r}$. Since $\frac{1}{3} < 1$, the series converges. Substitute the corresponding values for a and r in the formula for the series' sum, s .

$$\begin{aligned}s &= \frac{a}{1-r} \\&= \frac{1}{1 - (-1/3)} \\&= \frac{1}{4/3} \\&= \frac{3}{4}\end{aligned}$$